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EXPLORING THE USE OF BIOINSPIRED ROBOTS IN TREATING THE CHESAPEAKE BAY DEAD ZONE

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ABSTRACT

The Chesapeake Bay dead zone continues to harm local ecosystems and many species of marine life. While efforts have been taken to monitor the dead zone, a bioinspired robot could prove useful in collecting data. SeaTerp features an extensive sensing suite that can sample the same water for many tests, saving time and money.

NOMENCLATURE

V	VOLTS
PPM	PARTS PER MILLION

1. INTRODUCTION

The SeaTerp turtle-bot was built to address dead zones in the Chesapeake Bay. Dead zones are areas of water with little to no dissolved oxygen. Low oxygen levels cause psychological harm to marine life, can disrupt the food chain, and often lead to the deaths of hundreds of animals. [1] Dead zones form as a result of human pollution, most commonly from nitrogen and phosphorus found in wastewater runoff. This nitrogen and phosphorus promotes excess algae growth. Once the algae dies it is broken down by bacteria, a process which takes dissolved oxygen from nearby water. Algae blooms on the water's surface also block sunlight from reaching underwater vegetation, impacting animals that depend on the vegetation. [1]

According to the Chesapeake Bay Program, the Chesapeake Bay dead zone spans an average of 0.52 cubic miles in 2023. [2] While this is down from years past, the dead zone still inflicts considerable harm onto the Bay's ecosystem. Additionally, experts predict that climate change will only increase the size of the Chesapeake Bay dead zone. Increased rainfall and warmer temperatures are conducive for algae blooms, leading to dead zone expansion. [2] With this in mind, scientists have been researching possible solutions to the dead zone problem.

Researchers at the Chesapeake Bay Monitoring Program survey physical, chemical, and biological characteristics of the Chesapeake Bay dead zone. [3] Using this data, the researchers are able to predict changes in the dead zone and prescribe

solutions to policy makers. This project will assist in this effort, using the robot to map oxygen levels in and around the dead zone.

2. MATERIALS AND METHODS

The overall goal of the robot is to survey oxygen levels in the Chesapeake Bay. Inspiration was taken from the green sea turtle for their ability to swim efficiently, change directions with ease, and control buoyancy. [4] As part of its sensing suite, the robot would feature a hollow tube in its shell that would house a dissolved solids sensor and an exterior camera at the head. A crank-shaft system was used to actuate the robot's flippers forward and backward along an axis parallel to the robot's body. Propellers were placed at the rear of the robot to provide extra thrust. The locomotion design went through multiple iterations where the dimensions of the shell and base of the robot were altered to make it more hydrodynamic, pivoting from a round, football shape to a wider and flatter boat ballast shape. Lastly, a buoyancy bladder would allow the robot to sink and rise in the water.



FIGURE 1: CAD Model of SeaTerp's Flippers

An Arduino Uno was used to control the robot. The Arduino was on land and connected via tether to a breadboard inside the robot. Six independent DC motors were housed on board: 1 for each propeller, 1 for each flipper, and 2 for the buoyancy bladder. Two Adafruit Featherwing boards were used to control these motors, along with two 6V battery packs to power the Featherwings. LED lights and the dissolved solids sensor were plugged into the breadboard and derived their power from the computer on land. Finally, the camera plugged straight into the computer via USB. Arduino code was used to actuate the propellers, flippers, and bladder. An open source wiki page

provided Arduino code used to enable data collection using the dissolved solids sensor. [5]

Testing included above-water testing of the buoyancy bladder, dissolved solids sensor, flippers, and propellers. Ten tests of the bladder were conducted by taking in water from a half-filled cup and expelling the water to an empty cup. To test the dissolved solids sensor three cups were filled with water of differing quality: one with filtered drinking water, one with tap water, and one with coffee powder and salt mixed in. Five tests were conducted to see if the sensor could correctly detect pollutants in each cup. Two tests were conducted to see if the crankshaft system was able to actuate the flippers. Three tests were conducted of the propellers: one with both propellers, one with only the left propeller, and one with only the right propeller. Underwater testing also included tests of the propellers, buoyancy bladder, and dissolved solids sensor. These tests were conducted in the McKeldin Mall Fountain where, once in the water, each propeller was actuated individually then together. Next, the buoyancy bladder took in and expelled. Lastly, the robot took multiple samples of water to test.

3. RESULTS AND DISCUSSION

Preliminary above water testing of the buoyancy bladder was successful. The bladder was able to take in and expel an equal amount of water. Additionally the bladder proved to have minimal leakage. Above water testing of the dissolved solids sensor was also successful. When submerged in filtered drinking water, the sensor outputted low values (0-100 ppm). When submerged in tap water, the sensor outputted slightly higher values (500-1000 ppm). When submerged in the contaminated water, the sensor outputted very high values of over 1000 ppm. These results were across multiple tests. Above water testing of the flippers was a failure. The DC motors were unable to grip onto the wheels of the crankshaft system and thus were unable to actuate the flippers. The holes for the DC motors were slightly too large, and there was no fastening agent leading to slipping. As a result, the crankshaft system was scrapped in favor of static, non-moving flippers. Above water testing of the propellers had one failure and multiple successes. The only failure was when control of the propellers failed, most likely due to a loose connection in the wiring. Further testing yielded successful results, where each propeller could be controlled individually and in unison.

Final underwater tests of the buoyancy bladder were successful. As the bladder took in water the robot started to move down. Bubbles coming out of the robot as the robot expelled water showed that the system worked. Underwater testing of the dissolved solids sensor was inconclusive. Three tests of the sensor were conducted: one returned no value and two returned similar values. Since all the water in our testing environment was of the same quality, it is uncertain if the sensor returned an accurate value or some random value. Underwater testing of the propellers was successful. Each propeller was able to move

independently of the other and was able to turn the robot left and right. The propellers were also able to work in unison, moving the robot forward. After about a minute, the underwater test was cut short when water leaked into the robot through various gaps, disabling all the systems.

4. CONCLUSION

The bioinspired turtle robot SeaTerp successfully sampled water quality. The robot also featured 360° locomotion through the bladder and propeller system. By showing the capabilities of a diverse sensing suite and an integrated locomotion system, this research has contributed to bioinspired robotics research. In the future the crankshaft system could be improved upon to actuate the flippers. If done correctly this system can provide an incredible amount of power to propel the robot. More sensors, like pH or temperature sensors, could be added to our sensing suite. Housing more sensors on board would yield a more versatile robot. Lastly, with more time more caution would be taken in waterproofing the robot. This would uphold the integrity of the robot and its electronics allowing for longer tests and more data collection.

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